



ADC (F411 techref, pag. 212)

11.1 ADC introduction

The 12-bit ADC is a successive approximation analog-to-digital converter. It has up to 19 multiplexed channels allowing it to measure signals from 16 external sources, two internal sources, and the V_{BAT} channel. The A/D conversion of the channels can be performed in single, continuous, scan or discontinuous mode. The result of the ADC is stored into a left- or right-aligned 16-bit data register.

The analog watchdog feature allows the application to detect if the input voltage goes beyond the user-defined, higher or lower thresholds.





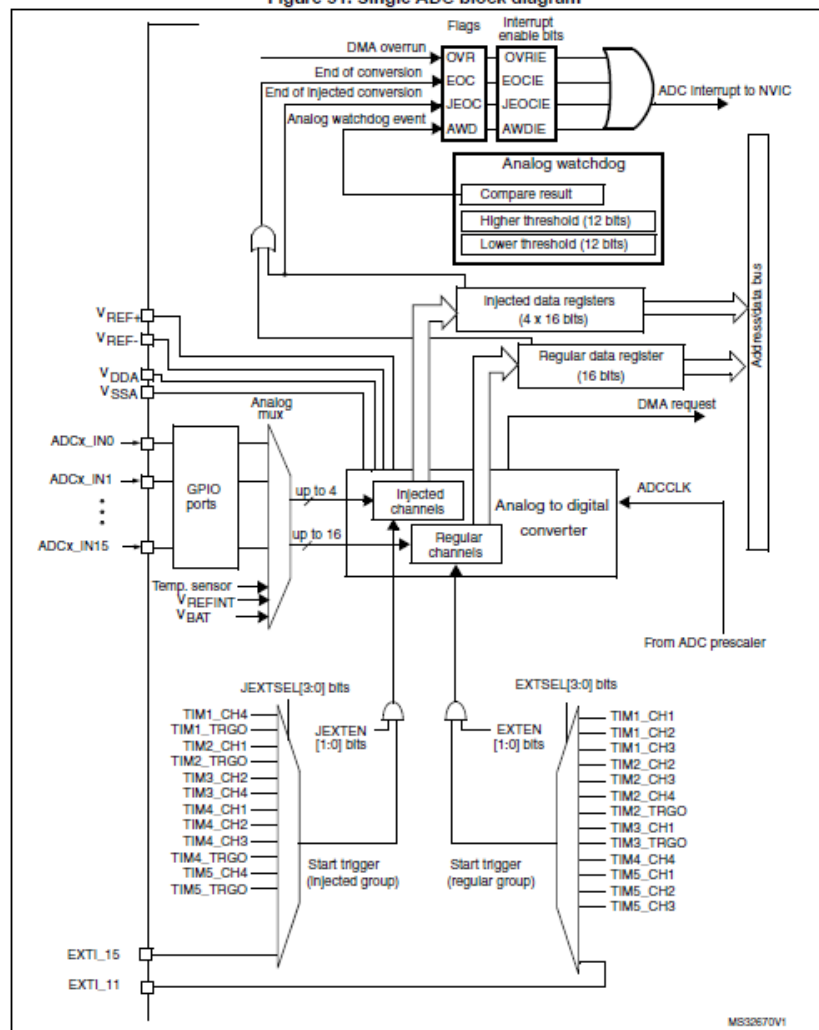
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11.2 ADC main features

- 12-bit, 10-bit, 8-bit or 6-bit configurable resolution
- Interrupt generation at the end of conversion, end of injected conversion, and in case of analog watchdog or overrun events
- Single and continuous conversion modes
- Scan mode for automatic conversion of channel 0 to channel 'n'
- Data alignment with in-built data coherency
- Channel-wise programmable sampling time
- External trigger option with configurable polarity for both regular and injected conversions
- Discontinuous mode
- ADC supply requirements: 2.4 V to 3.6 V at full speed and down to 1.8 V at slower speed
- ADC input range: $V_{REF-} \leq V_{IN} \leq V_{REF+}$
- DMA request generation during regular channel conversion

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Figure 31. Single ADC block diagram



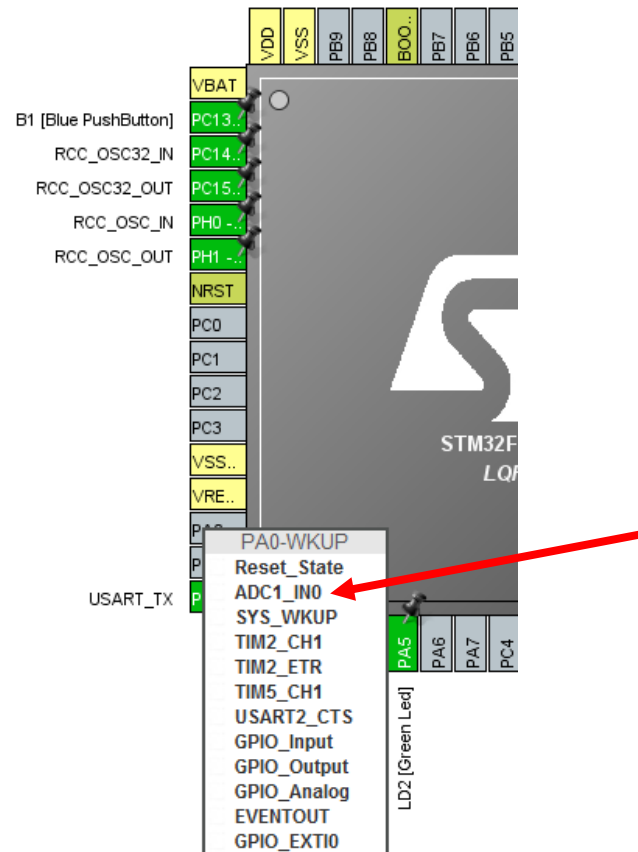


Lab Exercise:

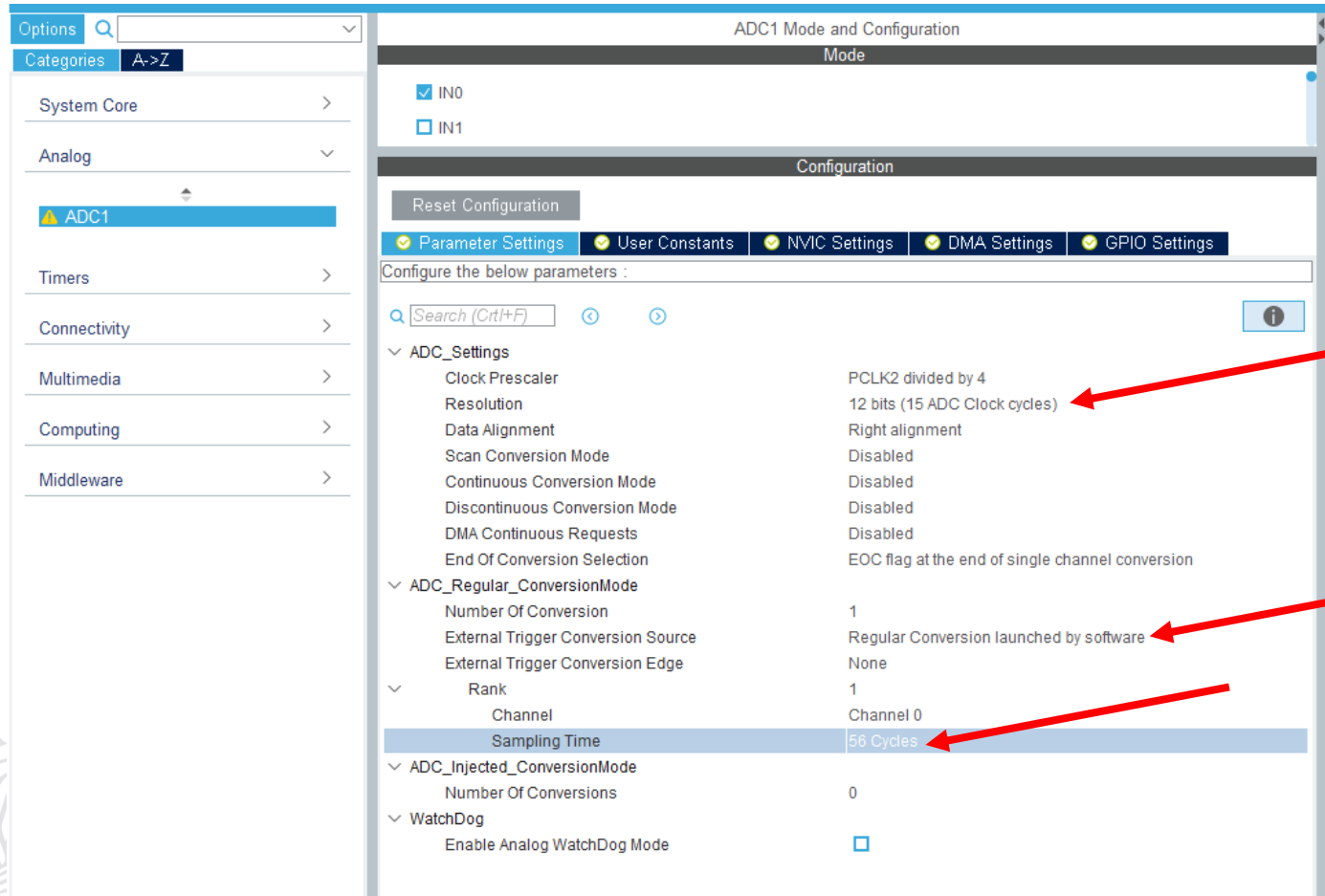
- Configure the ADC to sample PA0 on request
- Activate the ADC ISR
- Set PWM out (PA8, TIM1_CH1) to replicate the converted data.



Configure the ADC on PA0:



Configure the ADC:



ADC1 Mode and Configuration

Mode

- ☒ IN0
- ☐ IN1

Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | DMA Settings | GPIO Settings

Configure the below parameters :

Search (Ctrl+F)

- ADC_Settings**
 - Clock Prescaler: PCLK2 divided by 4
 - Resolution: 12 bits (15 ADC Clock cycles)
 - Data Alignment: Right alignment
 - Scan Conversion Mode: Disabled
 - Continuous Conversion Mode: Disabled
 - Discontinuous Conversion Mode: Disabled
 - DMA Continuous Requests: Disabled
 - End Of Conversion Selection: EOC flag at the end of single channel conversion
- ADC_Regular_ConversionMode**
 - Number Of Conversion: 1
 - External Trigger Conversion Source: Regular Conversion launched by software
 - External Trigger Conversion Edge: None
- Rank**
 - Rank: 1
 - Channel: Channel 0
- Sampling Time**: 56 Cycles
- ADC_Injected_ConversionMode**
 - Number Of Conversions: 0
- WatchDog**
 - Enable Analog WatchDog Mode: ☐



Activate the ISR:

Options
Categories **A-Z**

- System Core >
- Analog >
 - ADC1**
- Timers >
- Connectivity >

ADC1 Mode and Configuration

Mode

☒ IN0
☐ IN1

Configuration

Reset Configuration

☒ Parameter Settings
☒ User Constants
☒ **NVIC Settings**
☒ DMA Settings
☒ GPIO Settings

NVIC Interrupt Table	Enabled	Preemption Priority	Sub Priority
ADC1 global interrupt	☒	0	0





Modify the ISR:

```
void ADC_IRQHandler(void)
{
    /* USER CODE BEGIN ADC_IRQn 0 */

    /* USER CODE END ADC_IRQn 0 */
    HAL_ADC_IRQHandler(&hadc1);
    /* USER CODE BEGIN ADC_IRQn 1 */
    // something here to read from the ADC data reg (hadc1.Instance->DR)
    // and write to the PWM output register (htim1.Instance->CCR1)
    /* USER CODE END ADC_IRQn 1 */
}
```




Start the ADC from Main loop:

```
while (1)
{
    HAL_ADC_Start_IT(&hadc1);
    HAL_Delay(10);
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
}
```





Note that:

- Duty value (for TIM1_CH1) now updated in the ADC ISR.
- TIM1 period should be configured to 4096, to match the ADC conversion dynamic (12bit)
- Use >2kOhm resistor to feed the ADC pin from the function generator.
- Set the function generator to output 0-3.3V sinusoidal shape at 1Hz.
- Look with the oscilloscope at PA8.



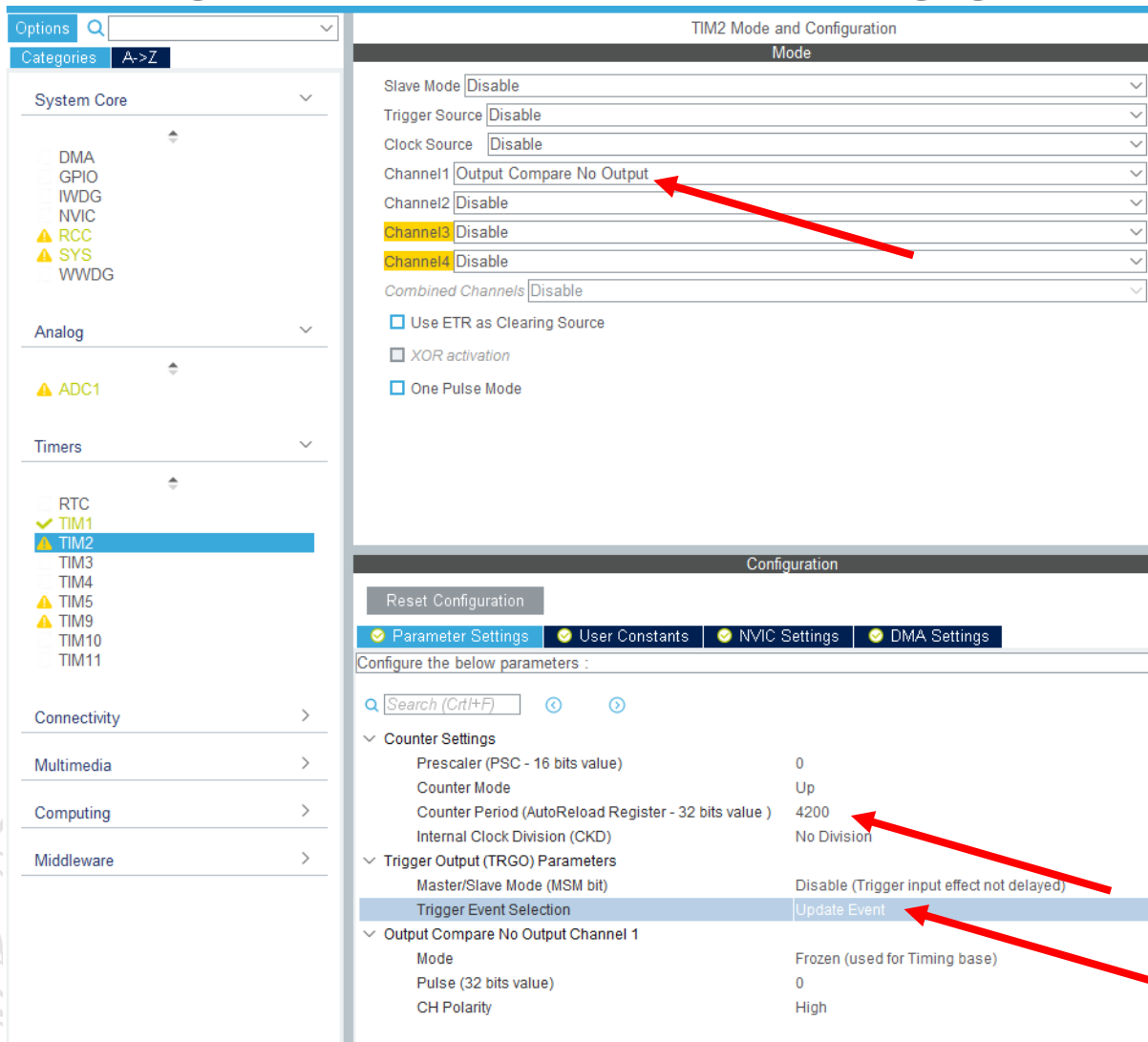


ADC automatic sampling:

- ADC can be configured to start sampling automatically when receives an 'event'
- We will use TIM2 to generate events to have ADC sampling data on a regular basis
- We will get PA0 values on the ADC ISR as usual



Configure TIM2 for trigger event:



Options

Categories **A-Z**

System Core

- DMA
- GPIO
- IWDG
- NVIC
- ▲ RCC
- ▲ SYS
- WWDG

Analog

- ▲ ADC1

Timers

- RTC
- ✓ TIM1
- ▲ TIM2
- TIM3
- TIM4
- ▲ TIM5
- ▲ TIM9
- TIM10
- TIM11

Connectivity >

Multimedia >

Computing >

Middleware >

TIM2 Mode and Configuration

Mode

Slave Mode

Trigger Source

Clock Source

Channel1

Channel2

Channel3

Channel4

Combined Channels

☐ Use ETR as Clearing Source

☐ XOR activation

☐ One Pulse Mode

Configuration

☒ Parameter Settings ☒ User Constants ☒ NVIC Settings ☒ DMA Settings

Configure the below parameters :

▼ **Counter Settings**

Prescaler (PSC - 16 bits value) 0

Counter Mode Up

Counter Period (AutoReload Register - 32 bits value) 4200

Internal Clock Division (CKD) No Division

▼ **Trigger Output (TRGO) Parameters**

Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)

Trigger Event Selection

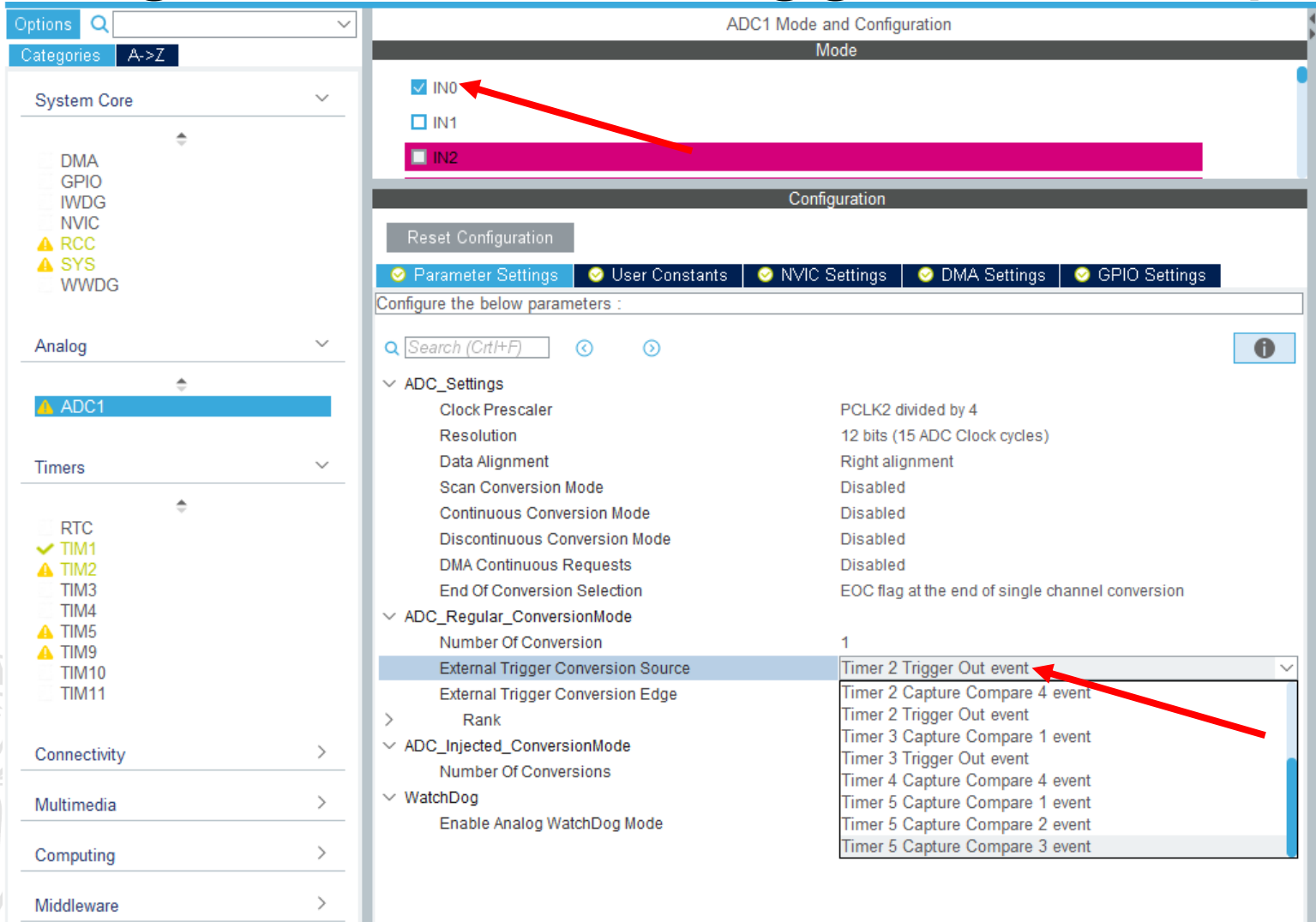
▼ **Output Compare No Output Channel 1**

Mode Frozen (used for Timing base)

Pulse (32 bits value) 0

CH Polarity High

Configure ADC for triggered sampling:



The screenshot shows the STM32CubeMX configuration interface for the ADC1. The left sidebar lists various system components, with 'ADC1' selected under the 'Analog' category. The main window displays the 'ADC1 Mode and Configuration' settings.

Mode:

- ☒ IN0
- ☐ IN1
- ☒ IN2

Configuration:

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | DMA Settings | GPIO Settings

Configure the below parameters :

Search (Ctrl+F)

ADC_Settings

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- Resolution: 12 bits (15 ADC Clock cycles)
- Data Alignment: Right alignment
- Scan Conversion Mode: Disabled
- Continuous Conversion Mode: Disabled
- Discontinuous Conversion Mode: Disabled
- DMA Continuous Requests: Disabled
- End Of Conversion Selection: EOC flag at the end of single channel conversion

ADC_Regular_ConversionMode

- Number Of Conversion: 1
- External Trigger Conversion Source: **Timer 2 Trigger Out event**
- External Trigger Conversion Edge: (dropdown)
- Rank: (dropdown)

ADC_Injected_ConversionMode

- Number Of Conversions: (dropdown)

WatchDog

- Enable Analog WatchDog Mode: (checkbox)



Start the system:

```
/* USER CODE BEGIN 2 */  
HAL_ADC_Start_IT(&hadc1);  
HAL_TIM_Base_Start(&htim2);  
HAL_TIM_Base_Start(&htim1);  
HAL_TIM_PWM_Start(&htim1,TIM_CHANNEL_1);
```

```
/* USER CODE END 2 */
```

```
/* Infinite loop */  
/* USER CODE BEGIN WHILE */  
while (1)  
{  
    /* USER CODE END WHILE */  
  
    /* USER CODE BEGIN 3 */  
}  
/* USER CODE END 3 */
```